```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
%matplotlib inline
warnings.filterwarnings('ignore')
pd.options.display.max_columns=99
```

```
df=pd.read_csv("/content/Dataset 1_origin.csv")
```

```
df.head()
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	MntFruits	MntMeatProducts	MntFishProducts	MntSweetProducts	MntGoldProds	NumDealsPurchases	NumWebPurchases	NumCatalogPurchases	NumStorePurchases	NumWebVisitsMonth	AcceptedCmp3	AcceptedCmp4	AcceptedCmp5	AcceptedCmp1	AcceptedCmp2	Complain	Z_CostContact	Z_Revenue	Response
0	5524	1957	Graduation	Single	58138.0	0	0	04-09-2012	58	635	88	546	172	88	88	3	8	10	4	7	0	0	0	0	0	0	3	11	1
1	2174	1954	Graduation	Single	46344.0	1	1	08-03-2014	38	11	1	6	2	1	6	2	1	1	2	5	0	0	0	0	0	0	3	11	0
2	4141	1965	Graduation	Together	71613.0	0	0	21-08-2013	26	426	49	127	111	21	42	1	8	2	10	4	0	0	0	0	0	0	3	11	0
3	6182	1984	Graduation	Together	26646.0	1	0	10-02-2014	26	11	4	20	10	3	5	2	2	0	4	6	0	0	0	0	0	0	3	11	0
4	5324	1981	PhD	Married	58293.0	1	0	19-01-2014	94	173	43	118	46	27	15	5	5	3	6	5	0	0	0	0	0	0	3	11	0

```
df.shape
```

```
(2240, 29)
```

```
df.tail()
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	MntFruits	MntMeatProducts	MntFishProducts	MntSweetProducts	MntGoldProds	NumDealsPurchases	NumWebPurchases	NumCatalogPurchases	NumStorePurchases	NumWebVisitsMonth	AcceptedCmp3	AcceptedCmp4	AcceptedCmp5	AcceptedCmp1	AcceptedCmp2	Complain	Z_CostContact	Z_Revenue	Response
2235	10870	1967	Graduation	Married	61223.0	0	1	13-06-2013	46	709	43	182	42	118	247	2	9	3	4	5	0	0	0	0	0	0	3	11	0
2236	4001	1946	PhD	Together	64014.0	2	1	10-06-2014	56	406	0	30	0	0	8	7	8	2	5	7	0	0	0	1	0	0	3	11	0
2237	7270	1981	Graduation	Divorced	56981.0	0	0	25-01-2014	91	908	48	217	32	12	24	1	2	3	13	6	0	1	0	0	0	0	3	11	0
2238	8235	1956	Master	Together	69245.0	0	1	24-01-2014	8	428	30	214	80	30	61	2	6	5	10	3	0	0	0	0	0	0	3	11	0
2239	9405	1954	PhD	Married	52869.0	1	1	15-10-2012	40	84	3	61	2	1	21	3	3	1	4	7	0	0	0	0	0	0	3	11	1

```
df.describe()
```

	ID	Year_Birth	Income	Kidhome	Teenhome	Recency	MntWines	MntFruits	MntMeatProducts	MntFishProducts	MntSweetProducts	MntGoldProds	NumDealsPurchases	NumWebPurchases	NumCatalogPurchases	NumStorePurchases	NumWebVisitsMonth	AcceptedCmp3	AcceptedCmp4	AcceptedCmp5	AcceptedCmp1	AcceptedCmp2	Complain	Z_CostContact	Z_Revenue	Response
count	2240.000000	2240.000000	2216.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.000000	2240.0	2240.0	2240.000000
mean	5592.159821	1968.805804	52247.251354	0.444196	0.506250	49.109375	303.935714	26.302232	166.950000	37.525446	27.062946	44.021875	2.325000	4.084821	2.662054	5.790179	5.316518	0.072768	0.074554	0.072768	0.064286	0.013393	0.009375	3.0	11.0	0.149107
std	3246.662198	11.984069	25173.076661	0.538398	0.544538	28.962453	336.597393	39.773434	225.715373	54.628979	41.280498	52.167439	1.932238	2.778714	2.923101	3.250958	2.426645	0.259813	0.262728	0.259813	0.245316	0.114976	0.096391	0.0	0.0	0.356274
min	0.000000	1893.000000	1730.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	3.0	11.0	0.000000
25%	2828.250000	1959.000000	35303.000000	0.000000	0.000000	24.000000	23.750000	1.000000	16.000000	3.000000	1.000000	9.000000	1.000000	2.000000	0.000000	3.000000	3.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	3.0	11.0	0.000000
50%	5458.500000	1970.000000	51381.500000	0.000000	0.000000	49.000000	173.500000	8.000000	67.000000	12.000000	8.000000	24.000000	2.000000	4.000000	2.000000	5.000000	6.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	3.0	11.0	0.000000
75%	8427.750000	1977.000000	68522.000000	1.000000	1.000000	74.000000	504.250000	33.000000	232.000000	50.000000	33.000000	56.000000	3.000000	6.000000	4.000000	8.000000	7.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	3.0	11.0	0.000000
max	11191.000000	1996.000000	666666.000000	2.000000	2.000000	99.000000	1493.000000	199.000000	1725.000000	259.000000	263.000000	362.000000	15.000000	27.000000	28.000000	13.000000	20.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	3.0	11.0	1.000000

```
df.isnull().sum()
```

	0
ID	0
Year_Birth	0
Education	0
Marital_Status	0
Income	24
Kidhome	0
Teenhome	0
Dt_Customer	0
Recency	0
MntWines	0
MntFruits	0
MntMeatProducts	0
MntFishProducts	0
MntSweetProducts	0
MntGoldProds	0
NumDealsPurchases	0
NumWebPurchases	0
NumCatalogPurchases	0
NumStorePurchases	0
NumWebVisitsMonth	0
AcceptedCmp3	0
AcceptedCmp4	0
AcceptedCmp5	0
AcceptedCmp1	0
AcceptedCmp2	0
Complain	0
Z_CostContact	0
Z_Revenue	0
Response	0

dtype: int64

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2240 entries, 0 to 2239
Data columns (total 29 columns):
 #   Column               Non-Null Count  Dtype  
---  ------               --------------  -----  
 0   ID                   2240 non-null   int64  
 1   Year_Birth           2240 non-null   int64  
 2   Education            2240 non-null   object 
 3   Marital_Status       2240 non-null   object 
 4   Income               2216 non-null   float64
 5   Kidhome              2240 non-null   int64  
 6   Teenhome             2240 non-null   int64  
 7   Dt_Customer          2240 non-null   object 
 8   Recency              2240 non-null   int64  
 9   MntWines             2240 non-null   int64  
 10  MntFruits            2240 non-null   int64  
 11  MntMeatProducts      2240 non-null   int64  
 12  MntFishProducts      2240 non-null   int64  
 13  MntSweetProducts     2240 non-null   int64  
 14  MntGoldProds         2240 non-null   int64  
 15  NumDealsPurchases    2240 non-null   int64  
 16  NumWebPurchases      2240 non-null   int64  
 17  NumCatalogPurchases  2240 non-null   int64  
 18  NumStorePurchases    2240 non-null   int64  
 19  NumWebVisitsMonth    2240 non-null   int64  
 20  AcceptedCmp3         2240 non-null   int64  
 21  AcceptedCmp4         2240 non-null   int64  
 22  AcceptedCmp5         2240 non-null   int64  
 23  AcceptedCmp1         2240 non-null   int64  
 24  AcceptedCmp2         2240 non-null   int64  
 25  Complain             2240 non-null   int64  
 26  Z_CostContact        2240 non-null   int64  
 27  Z_Revenue            2240 non-null   int64  
 28  Response             2240 non-null   int64  
dtypes: float64(1), int64(25), object(3)
memory usage: 507.6+ KB
```

```
plt.style.use("fivethirtyeight")
```

```
x_question=df.iloc[:,5:33]
x_question = x_question.select_dtypes(include=np.number) # Select only numeric columns
q_mean=x_question.mean(axis=0)
total_mean=q_mean.mean()
```

```
q_mean.reset_index(level=0,inplace=True)
```

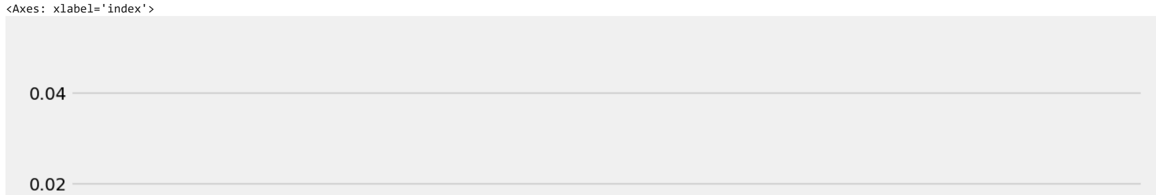
```
q_mean.head()
```

	level_0	index	mean
0	0	Kidhome	0.444196
1	1	Teenhome	0.506250
2	2	Recency	49.109375
3	3	MntWines	303.935714
4	4	MntFruits	26.302232

Next steps: Generate code with q_mean New interactive sheet

```
plt.figure(figsize=(14,7))
sns.barplot(x="index",y=0,data=q_mean)
```

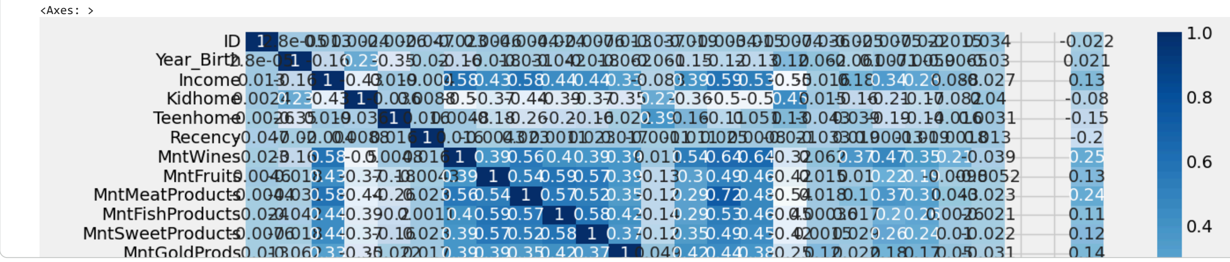
```
<Axes: xlabel='index'>
```




```
corr=df.select_dtypes(include=np.number).corr()
```

```
plt.figure(figsize=(14,7))
sns.heatmap(corr,annot=True,cmap="Blues")
```

```
<Axes: >
```




```
x=df.iloc[:,5:33]
```

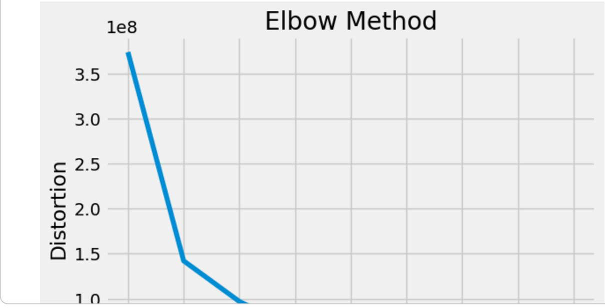
```
from sklearn.decomposition import PCA
pca=PCA(n_components=2,random_state=42)
x_numeric = x.select_dtypes(include=np.number) # Filter for numeric columns
x_pca=pca.fit_transform(x_numeric)
```

```
x_pca
pca.explained_variance_ratio_
```

```
array([0.78671237, 0.17131975])
```

```
from sklearn.cluster import KMeans
distortion=[]
cluster_range=range(1,10)
for i in cluster_range:
  model=KMeans(n_clusters=i,init="k-means++",random_state=42)
  model.fit(x_pca)
  distortion.append(model.inertia_)
```

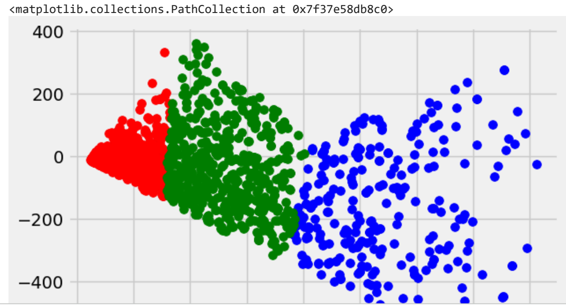
```
plt.plot(cluster_range,distortion)
plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("Distortion")
plt.show()
```




```
model=KMeans(n_clusters=4,init="k-means++",random_state=42)
model.fit(x_pca)
y=model.predict(x_pca)
```

```
plt.scatter(x_pca[y==0,0],x_pca[y==0,1],s=50,c="red",label="Cluster 1")
plt.scatter(x_pca[y==1,0],x_pca[y==1,1],s=50,c="blue",label="Cluster 2")
plt.scatter(x_pca[y==2,0],x_pca[y==2,1],s=50,c="green",label="Cluster 3")
```

```
<matplotlib.collections.PathCollection at 0x7f37e58db8c0>
```




```
plt.scatter(model.cluster_centers_[:,0],model.cluster_c
plt.title("Clusters of Customers")
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")
plt.legend()
plt.show()
```